

Vulnerability Assessment Report

Task 5 — OpenVAS Automated Scanning

Execution Attempt & Root-Cause Analysis of an Incomplete Feed Synchronization

ENGAGEMENT

CYS 4151 Ethical Hacking — Final Examination, Setting A (Summer 2026)

DELIVERABLE

Task 5 — Vulnerability Assessment (6 marks)

ASSESSMENT NODE

PC5 — OpenVAS-on-Kali scanner (192.168.50.250)

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TEAM ROLE

Blue Team — OpenVAS vulnerability scanning & CVSS triage

DATE OF TESTING

03 October 2026

REPORT STATUS

Incomplete — assessment blocked (see §7, Root-Cause Analysis)

CLASSIFICATION

Confidential — Academic Lab Use Only

1 Executive Summary

Task 5 of the SecureTech Solutions security assessment required an automated vulnerability scan of the lab target set (192.168.50.20, .30 and .40) using OpenVAS / Greenbone Vulnerability Management (GVM), followed by manual cross-checking and CVSS-based triage of at least five findings. This report documents that the task could not be completed as specified.

Although the OpenVAS service and its web interface were operational, the Greenbone data feeds that GVM depends on had not finished synchronizing. The SCAP data feed remained in an “Update in progress” state and the GVM Data feed — the component that supplies the default Port Lists and scan configurations — had not been imported. As a direct result, the scanner refused to create a scan target, reporting that the default Port List was unavailable, and no scan task could therefore be created or launched. No vulnerability results were produced.

The synchronization was attempted five times. On the fifth and final attempt observed, the SCAP data store grew from approximately **1 001 MB to 1.1 GB over roughly one hour** before stalling without completing. Substantially increasing the VM’s resources (disk, memory and processors, detailed in §9) did not change the outcome. The failure is an environmental / configuration blocker, not a finding about the target network. This report records the evidence, establishes the root cause, assesses the impact on the wider engagement, and sets out the remediation required to bring the feeds to a “Current” state so that Task 5 can be re-run.

2 Objective and Scope

Per the CYS 4151 execution guide, Task 5 is owned by PC5 (Amina) on the OpenVAS-on-Kali host and calls for the following:

- Start the GVM / OpenVAS stack and reach the web UI at `https://192.168.50.250:9392`.
- Create a target list covering 192.168.50.20, 192.168.50.30 and 192.168.50.40.
- Launch a “Full and Fast” scan against that target set.
- Export the results sorted by CVSS severity and document at least five vulnerabilities.
- Cross-check at least two findings manually (`searchsploit`, `nmap --script vuln`).

Scope of this report: the vulnerability-scanning activity performed from PC5 only. It does not cover the Red Team tasks (2, 3, 6–10) or the other Blue Team tasks, except where the failure of Task 5 affects them (see §8).

3 Test Environment

- **Scanner host:** Kali Linux VM (`amina@vbox`) running under Oracle VirtualBox; OpenVAS / GVM with the Greenbone Security Assistant web UI reached locally at `127.0.0.1:9392` (published as 192.168.50.250 on the lab LAN).
- **Scanner VM resources:** during the OpenVAS / GVM installation an error was encountered and resolved by raising the VM’s base memory from 2000 MB to 4000 MB. Ahead of the Task 5 feed-sync attempts the VM was provisioned further still — the virtual disk expanded from 30 GB to 80 GB, the base memory raised to 8000 MB, and 2 processors allocated.
- **Target set:** lab hosts on 192.168.50.0/24 — Rayan’s server (.20), the nested DVWA / Juice Shop host (.30) and Metasploitable (.40); the host field also included additional lab addresses (... , .100, .200).

- **Feeds required by GVM:** NVT (Greenbone Community Feed), SCAP data (CVE / CPE), CERT data, and GVM Data (default Port Lists, scan configs, report formats).
- **Identity overlay:** each capture carries the required on-screen identity (name “Amina Boubakary” and Student ID ICTU20241870) and the live VirtualBox / system clock, per the lab’s evidence rules.

4 Methodology

The intended procedure followed the Blue Team strategy in the execution guide: run scanning proactively and independently, obtain breadth with a Full and Fast scan, then add depth with manual cross-checks. The sequence attempted was:

1. Start the GVM stack (`sudo gvm-start`) and open the web UI.
2. Confirm feed readiness via Administration → Feed Status.
3. Create the target “CYS4151-Targets” with the lab host list.
4. Create and launch a Full and Fast scan task against the target.
5. Export and triage the results by CVSS; cross-check with `searchsploit / nmap --script vuln`.

Steps 1–2 were reached; the process could not progress past step 3 because of the feed state described below. Feed progress was monitored from the command line with:

```
watch sudo du -sh /var/lib/gvm/scap-data/
```

5 Chronology of the Attempt

The session is summarised below in logical phases. The two command-line timestamps (12:59:58 PM and 01:26:53 PM EDT) are read directly from the monitoring output; the remaining phases occurred within the same working session on 03 October 2026.

Phase	What was observed	Evidence
A — Feed check	Administration → Feed Status reviewed. The NVT feed (Greenbone Community Feed) read “Current”, but the SCAP data feed read “Update in progress...”. The default Port Lists and scan configurations were therefore not yet available.	Figure 1
B — Sync growing	The SCAP data store was monitored from the shell. It stood at $\approx 1\,001$ MB (1002M) at 12:59:58 PM EDT and was still importing.	Figure 2
C — gvmd warnings	gvmd logged repeated warnings that it could not read the GVMD Data and CERT feed metadata (feed.xml “No such file or directory”) — confirming those feeds had not yet been imported.	Figure 3
D — Still incomplete	On the 5th synchronization attempt the SCAP store reached 1.1 GB at 01:26:53 PM EDT and continued for roughly one hour without ever reaching “Current”.	Figure 4
E — Target config	The “CYS4151-Targets” target was configured in the New Target dialog with the lab host list (192.168.50.30, .40, .100, .200, ...).	Figure 5
F — Blocked on save	Saving the target was rejected: “the default Port List is not available ... the feed not having completed its synchronization.” No scan task could then be created or launched.	Figure 6

6 Observations and Evidence

All screenshots were captured on the PC5 scanner VM during the Task 5 attempt and carry the required name / Student ID / live-clock overlay.

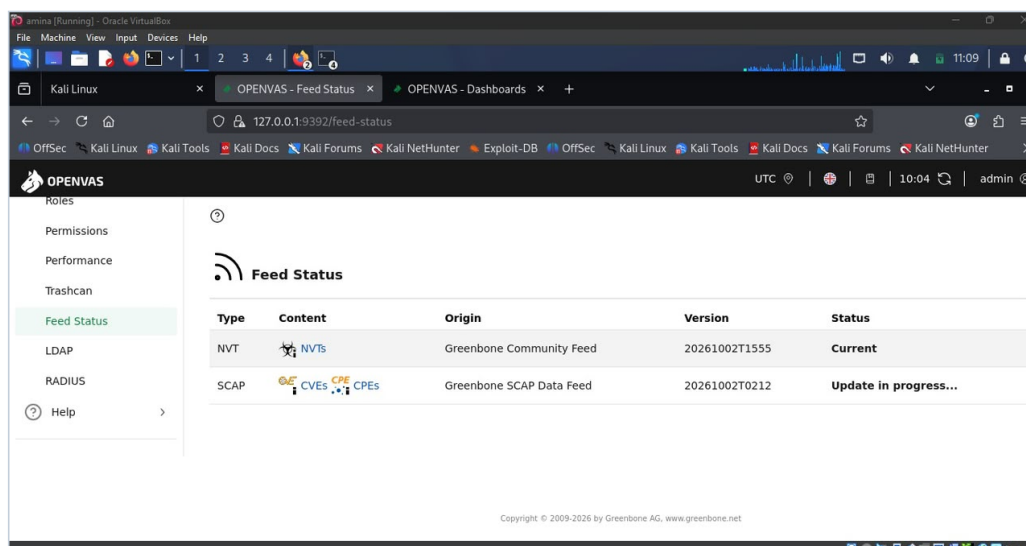


Figure 1. Feed Status. The NVT feed reads “Current”, but the Greenbone SCAP Data Feed reads “Update in progress...”. Until SCAP (and the GVMD Data feed) finish, the scanner has no default Port List or scan config to work with.

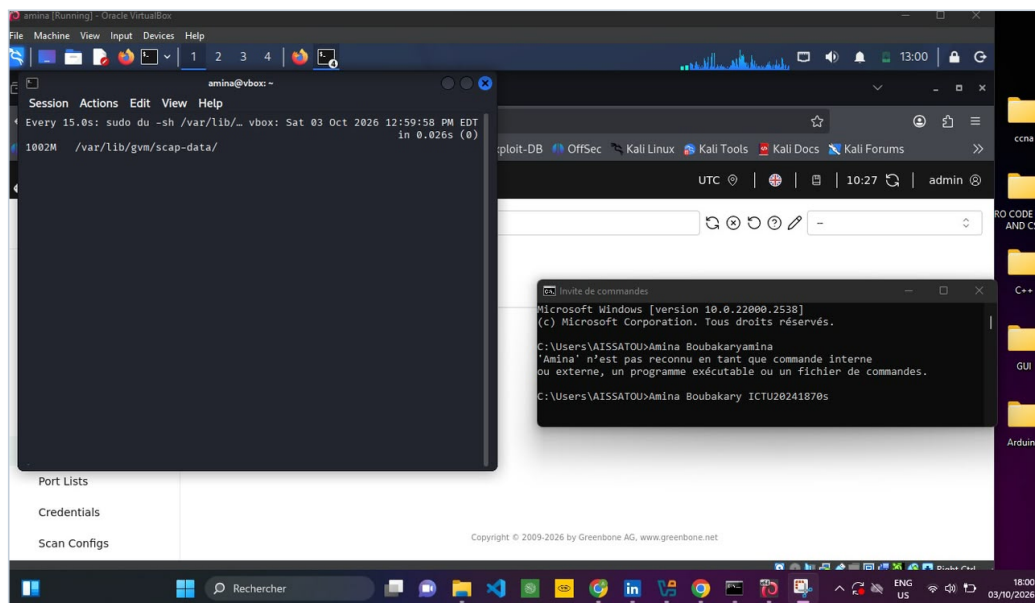


Figure 2. SCAP data store at $\approx 1\,001$ MB (1002M) at 12:59:58 PM EDT, still importing — monitored with `watch du -sh` on `/var/lib/gvm/scap-data/`.

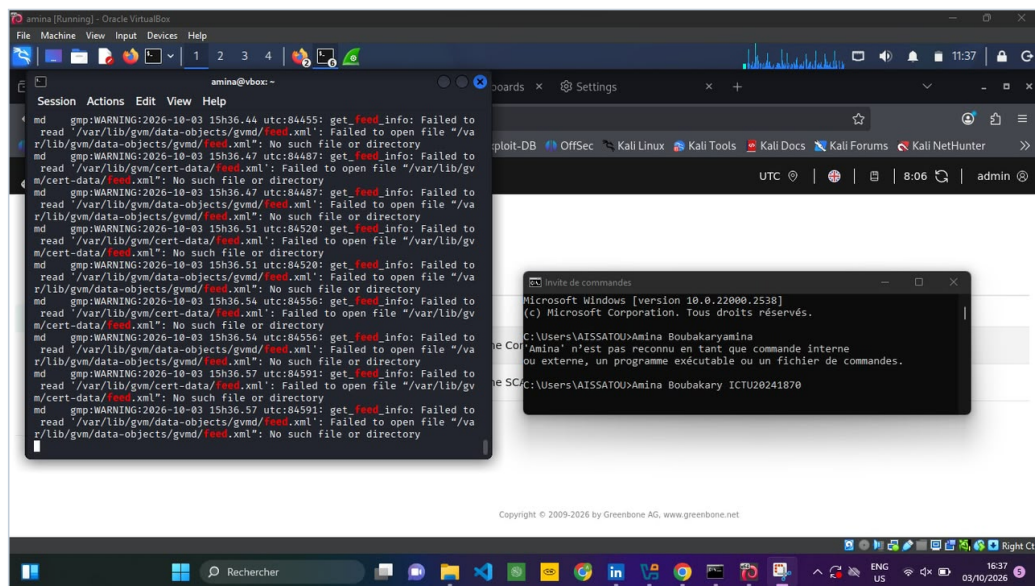


Figure 3. `gvmcd` warnings: `get_feed_info` repeatedly fails to read `/var/lib/gvm/data-objects/gvmd/feed.xml` and `/var/lib/gvm/cert-data/feed.xml` ("No such file or directory") — the GVMD Data and CERT feeds had not been synced.

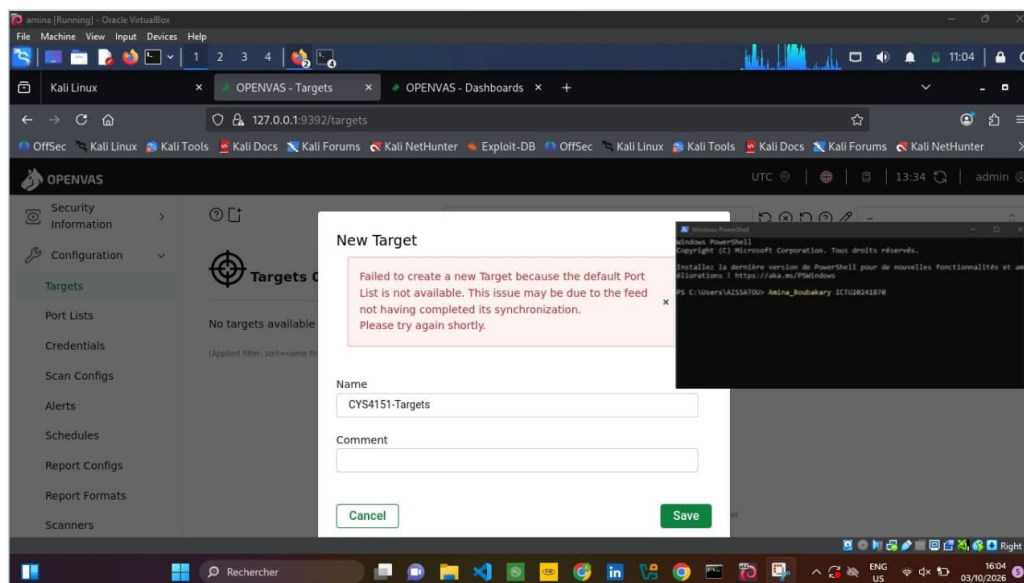


Figure 6. Target creation rejected: “Failed to create a new Target because the default Port List is not available. This issue may be due to the feed not having completed its synchronization.” With no target, no scan task could be created.

7 Root-Cause Analysis

GVM cannot scan until several locally-held data feeds are synchronized and imported: the NVT feed (the vulnerability tests), the SCAP data feed (CVE / CPE data loaded into the gvmdb database), the CERT data feed, and the GVMDB Data feed, which provides the default Port Lists, the “Full and Fast” scan config and the report formats.

Primary cause — incomplete feed synchronization. The SCAP feed was still importing (“Update in progress”, Figure 1) and the GVMDB Data feed had not loaded, shown directly by gvmdb’s inability to read its `feed.xml` (Figure 3). Because the GVMDB Data feed is exactly what supplies the default Port List, its absence produced the precise error seen on save (Figure 6). The scanner was installed correctly, but it had nothing to build a target or a scan config from.

Contributing factors:

- **Feed volume:** the SCAP / CVE dataset is large (over 1 GB here), and the Greenbone Community Feed is rate-limited, so a full sync can take hours or stall on a constrained connection.
- **Repeated restarts:** five separate attempts can each interrupt and re-lock the import, so the sync never gets an uninterrupted window to finish.
- **Resources were increased but it still failed:** an earlier installation error on this host was cleared by raising the base memory from 2000 MB to 4000 MB, and for Task 5 the VM was pushed further (disk 30 GB to 80 GB, memory 8000 MB, 2 processors). The synchronization still did not complete — indicating the remaining bottleneck is the feed download / import itself (data volume and the rate-limited community feed), not local CPU, RAM or disk.
- **Partial-sync signature:** NVT reaching “Current” while SCAP and GVMDB Data lagged is consistent with the heavier CVE import simply not having finished.

Important distinction: this is an environmental / tooling blocker on the scanner, not a vulnerability or misconfiguration of the target network. It says nothing about the security posture of 192.168.50.20 / .30 / .40.

8 Impact on the Assessment

- **Task 5 deliverables not produced:** no target, no Full and Fast scan, no CVSS-sorted results, and none of the required five-plus documented vulnerabilities.
- **Downstream tasks affected:** Task 6 (exploitation) loses the scanner-confirmed finding it is meant to select from; Task 11 (Blue Team detection) loses the OpenVAS evidence source; Task 12 (hardening) loses the OpenVAS findings feeding its remediation list.
- **Partial mitigation available:** the guide's manual cross-checks (`searchsploit`, `nmap --script vuln`) do not depend on the GVM feeds and can surface interim findings while the feed issue is resolved.

9 Troubleshooting Performed

- Resolved an installation error hit while setting up OpenVAS / GVM by increasing the VM's base memory from 2000 MB to 4000 MB.
- Verified feed state in Administration → Feed Status (SCAP in progress, NVT current).
- Monitored the SCAP data directory growth with `watch sudo du -sh /var/lib/gvm/scap-data/` (1002 MB → 1.1 GB).
- Inspected `gvmd` output and identified the missing GVMD Data / CERT `feed.xml` warnings.
- Re-attempted target creation several times — the same default-Port-List error recurred each time.
- Re-ran the full synchronization up to a fifth attempt; the fifth run (1.1 GB) continued for about an hour without completing.
- Increased the VM's resources for the scan attempts — expanded the virtual disk from 30 GB to 80 GB, raised the base memory to 8000 MB, and allocated 2 processors — but the synchronization and scan were still unsuccessful.

10 Recommendations and Remediation

To bring the scanner to a working state and re-run Task 5:

1. **Run one uninterrupted sync to completion.** Avoid repeated restarts. Use `sudo greenbone-feed-sync`, which pulls the NVT, SCAP, CERT and GVMD Data feeds; allow several hours as the community feed is large and rate-limited.
2. **Confirm all feeds read “Current”** in Administration → Feed Status — not just NVT — before creating any target.
3. **Verify the GVMD Data feed actually loaded:** Configuration → Port Lists should list the defaults (e.g. “All IANA assigned TCP”) and Configuration → Scan Configs should show “Full and Fast”. Their absence is what caused the error.
4. **If a feed is stuck, sync components individually** to isolate it: `sudo greenbone-feed-sync --type scap` and `sudo greenbone-feed-sync --type gvmd-data`.
5. **After syncing, rebuild and restart:** `sudo gvmd --rebuild`, then `sudo gvm-stop && sudo gvm-start` (or restart the `gvmd` and `osspd-openvas` services).
6. **Run** `sudo gvm-check-setup` to confirm that no feed component is empty or missing.
7. **Resources are already ample — focus on time and connectivity:** with the disk at 80 GB,

8000 MB RAM and 2 processors, the outstanding requirement is an uninterrupted, stable connection; schedule the sync at off-peak hours and let it run to completion before the lab session. When enlarging the virtual disk, confirm with `df -h` inside Kali that the added space is actually available, since a VirtualBox disk resize does not grow the guest partition on its own.

8. **Once Feed Status is fully “Current”**, re-run Task 5 from target creation through the Full and Fast scan and the CVSS export.

11 Conclusion

The OpenVAS vulnerability assessment for Task 5 could not be completed because the Greenbone data feeds — specifically the SCAP data feed and the GVM Data feed that supplies the default Port Lists and scan configurations — never finished synchronizing, despite five attempts, a final run of roughly one hour at 1.1 GB, and substantial increases to the VM’s disk (30 GB to 80 GB), memory (to 8000 MB) and processors (to 2). The scanner itself was installed and reachable; the blocker is environmental — the feed download and import, not local resources. With the feeds brought fully to “Current” and the service rebuilt and restarted as set out in §10, the target can be created and the Full and Fast scan executed, allowing Task 5 to be delivered as specified. Until then, the manual cross-checks remain available to produce interim findings.

Appendix A — Evidence Inventory

All captures taken on the PC5 OpenVAS-on-Kali VM, 03 October 2026, with the name / Student ID / live-clock overlay visible.

- Figure 1 — Feed Status: SCAP “Update in progress”, NVT “Current”.
- Figure 2 — SCAP data store at $\approx$ 1 001 MB (12:59:58 PM EDT).
- Figure 3 — gvmfeed.xml read failures (GVMD Data / CERT not synced).
- Figure 4 — SCAP data store at 1.1 GB (01:26:53 PM EDT), 5th attempt.
- Figure 5 — New Target dialog with the lab host list.
- Figure 6 — Target creation blocked: default Port List not available.

Commands referenced

```
sudo gvm-start
```

```
watch sudo du -sh /var/lib/gvm/scap-data/
```

```
sudo greenbone-feed-sync          # all feeds
```

```
sudo greenbone-feed-sync -type scap
sudo greenbone-feed-sync -type gvm-data
```

```
sudo gvm -rebuild
sudo gvm-stop && sudo gvm-start
```

```
sudo gvm-check-setup
```